

3D Brain Tumor Segmentation

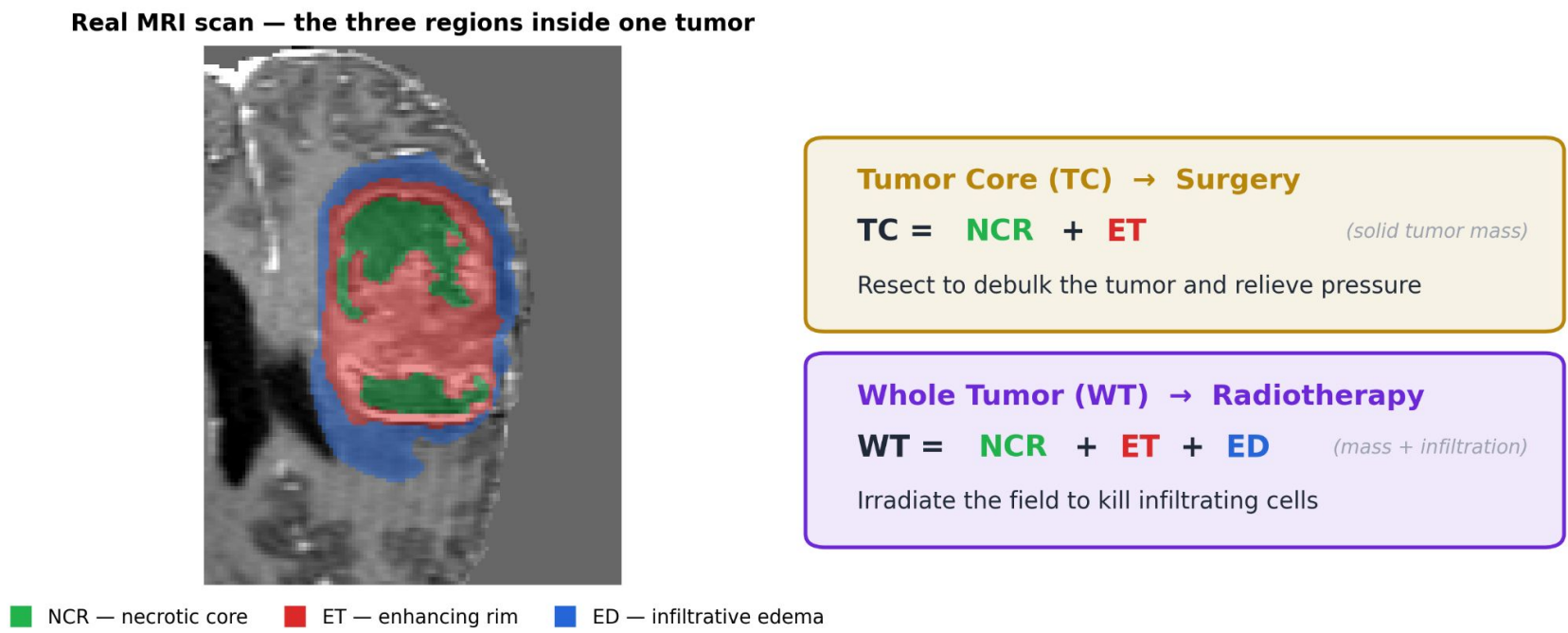
AURAS — A Hybrid CNN–Transformer Architecture

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1. Clinical Problem

- Gliomas are aggressive brain tumors. Doctors assess them from three nested regions, each guiding a different decision.
- Whole Tumor (WT) — full extent, used for surgical and radiotherapy planning
 - Tumor Core (TC) — the resectable mass, guides resection boundaries
 - Enhancing Tumor (ET) — active tumor tissue, the key marker of aggressiveness

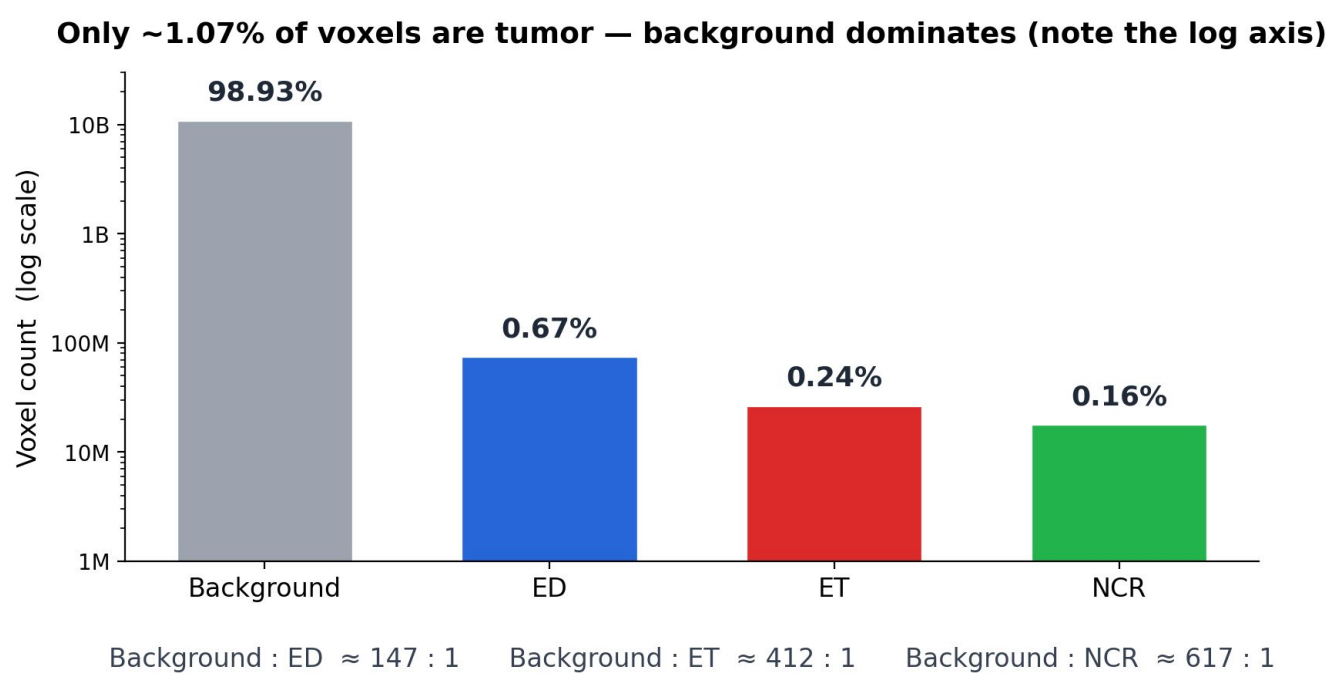
Hand-labeling these across 4 MRI scans is slow and inconsistent — and a useful model should also flag when it is unsure.



2. Data & Challenge

- BraTS 2021 — 1251 multimodal MRI volumes (1000 train / 251 val) with labels.
- 4 MRI modalities per case: T1, T1CE, T2, FLAIR
 - T1CE highlights ET; FLAIR shows WT extent
 - Stacked into a 5-channel 128³ input

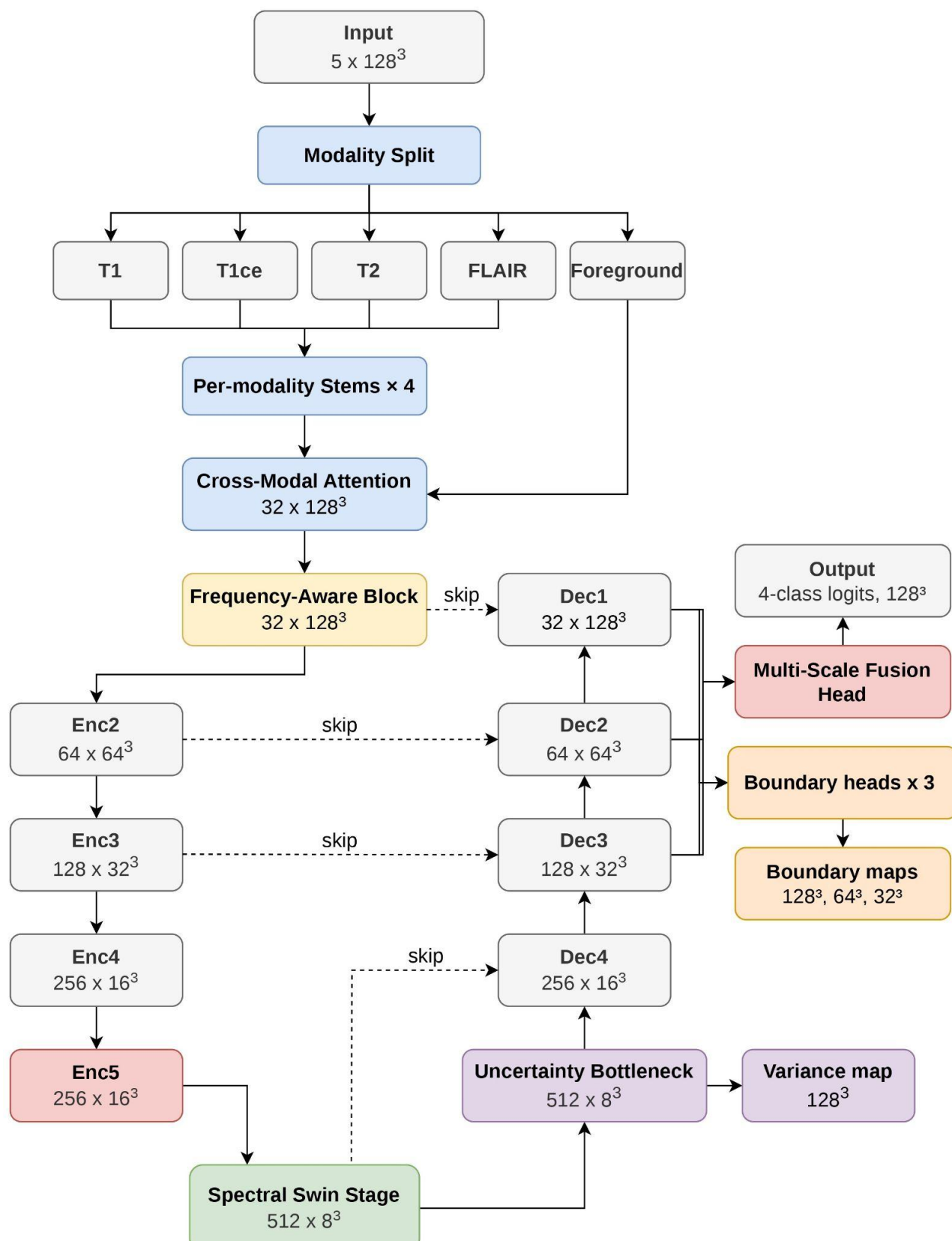
Tumor voxels are rare and ET is the smallest, so the loss and patch sampling are weighted toward the tumor.



3. AURAS Architecture

AURAS is one configurable 3D Trans-ResU-Net: a residual U-Net backbone enhanced with attention and spectral modules to capture both fine detail and global context.

- Modality stems + cross-modal attention — smartly fuse the 4 MRIs
- Frequency + Spectral-Swin — add global context the CNN alone misses
- Predictive-variance head — outputs a per-voxel uncertainty map
- Boundary head — sharpens tumor edges



4. Training & Loss

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{seg}}}_{\text{accuracy}} + \underbrace{\lambda_{\text{unc}} |\bar{v} - t|}_{\text{trust}} + \underbrace{\lambda_b(t) \mathcal{L}_{\text{bnd}}}_{\text{edges}}$$

$$\mathcal{L}_{\text{seg}} = \sum_{s \in \{1, \frac{1}{2}, \frac{1}{4}\}} w_s \sum_{r \in \{\text{WT}, \text{TC}, \text{ET}\}} (\text{Dice}_r + \text{Focal}_r)$$

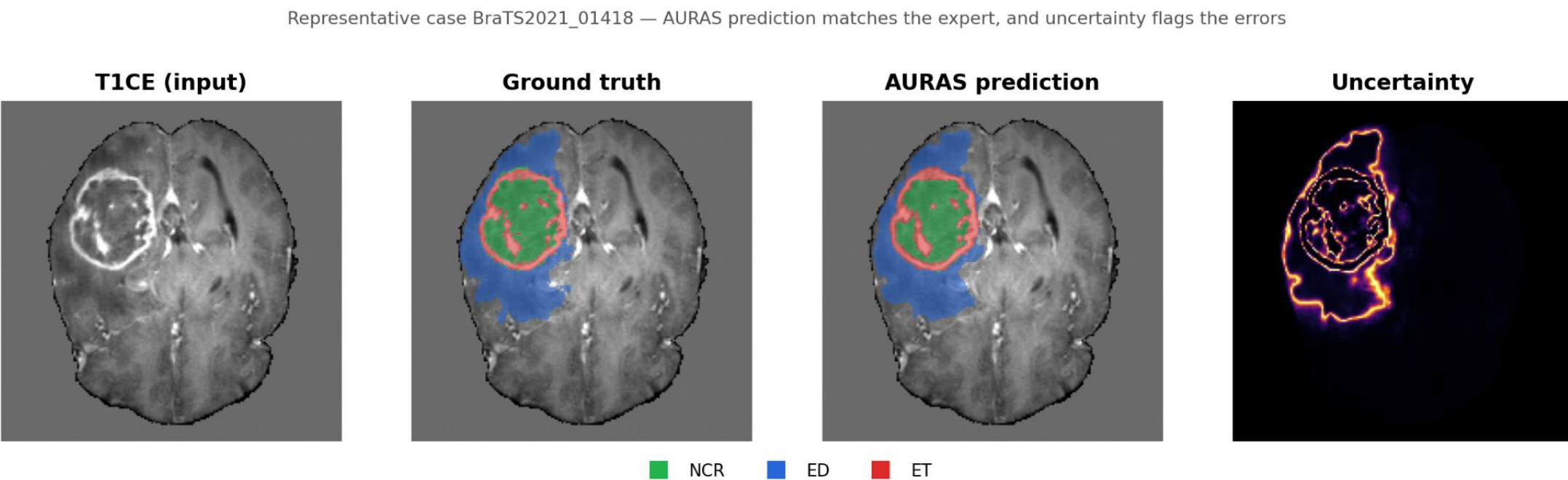
$$\text{Dice}_r = 1 - \frac{2 \sum p_r g_r}{\sum p_r + \sum g_r} \quad \text{Focal}_r = - \sum (1 - p_t)^\gamma \log p_t$$

The model is trained directly on the clinical regions (WT / TC / ET), so the objective matches how it is evaluated.

- Region-wise Dice + Focal loss — handles imbalance and hard voxels
- Deep supervision — stabilizes early training
- Uncertainty + boundary terms — added gradually during training
- EMA weights, snapshot ensemble, and flip-TTA at inference

5. Qualitative Result

A representative validation case: input MRI, expert ground truth, and the AURAS prediction. AURAS recovers fine boundaries and the small enhancing core that simpler models miss.



6. Headline Results

Metric	3D U-Net	SegResNet	SwinUNETR	Base CNN	AURAS
Dice WT				0.923	0.929
Dice TC				0.788	0.781
Dice ET				0.765	0.789
HD95 (mm) ↓				6.75	5.95
NSD ↑				0.787	0.803

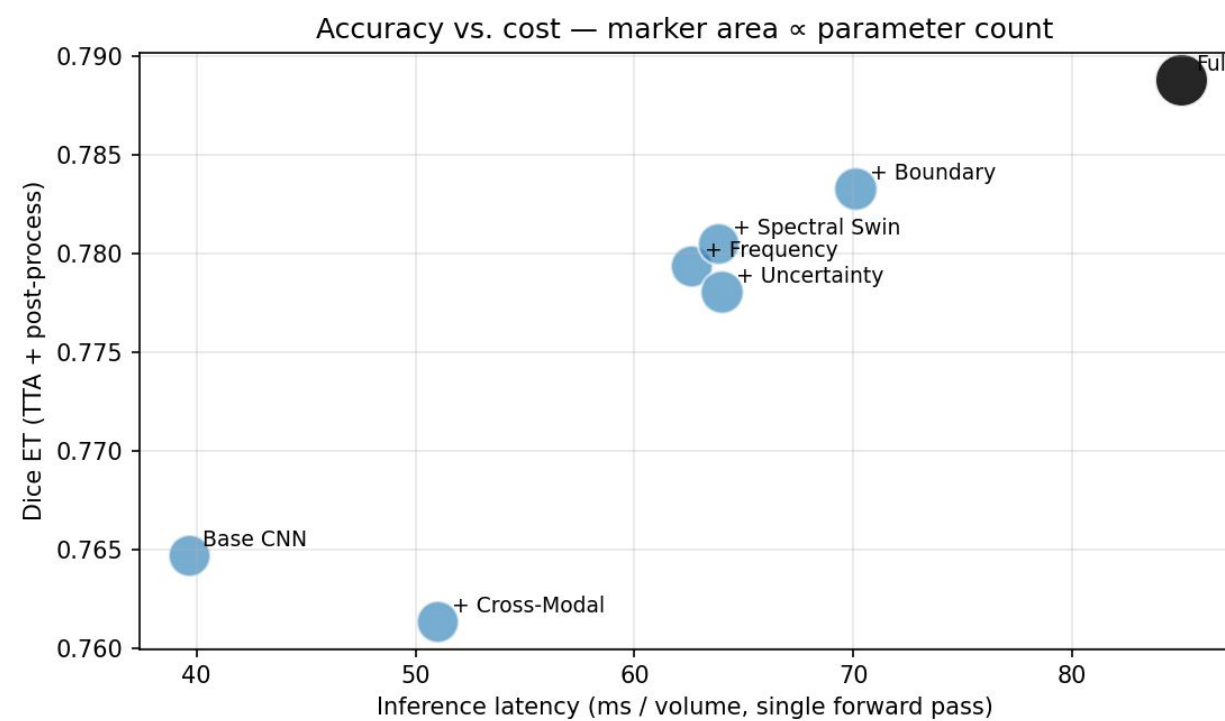
Evaluated on the 251-case validation split with test-time augmentation and post-processing.

- AURAS improves Whole Tumor Dice and boundary accuracy (HD95) over the Base CNN, with statistical significance (paired Wilcoxon, Bonferroni-corrected p < 0.01)
- It also lifts the hardest region, Enhancing Tumor, by ~2 Dice points
- Comparison against SwinUNETR / SegResNet / 3D U-Net is in progress

7. Efficiency

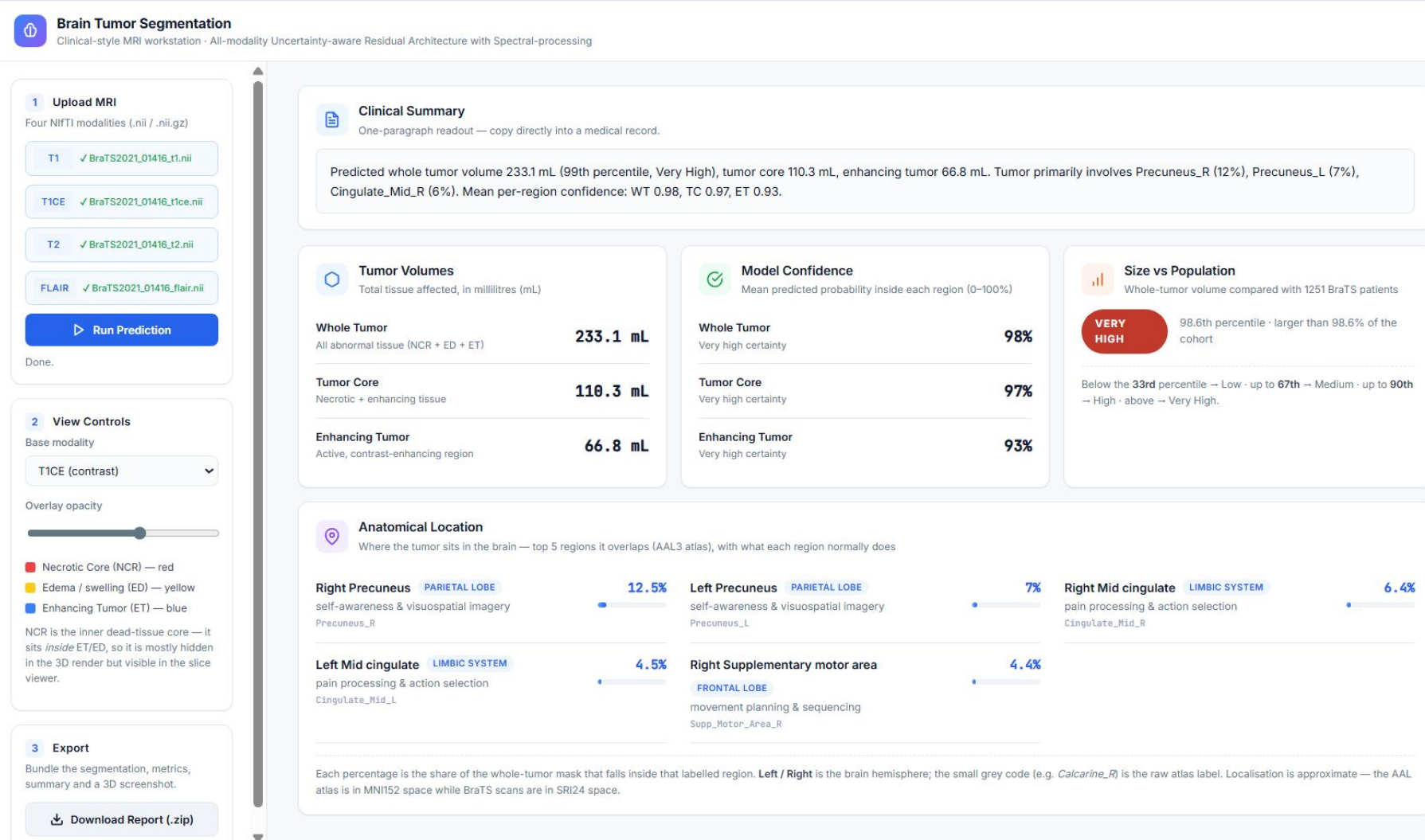
Accuracy comes at a moderate, deployable cost.

- 37 M parameters
- ~85 ms per volume on a single GPU
- ~2.4 GB VRAM at 128³ inference



8. Deployment

AURAS is deployed in an interactive web workstation: it loads a study, predicts the three tumor regions, and reports a per-region confidence score for clinical review.



9. Conclusion

A single hybrid CNN–Transformer model delivers accurate, uncertainty-aware brain tumor segmentation — improving whole-tumor accuracy and boundary quality over a strong CNN baseline, and deployable in an interactive clinical workstation.